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Brazing of SiC ceramic with extraterrestrial regolith simulant in ISRU and ISRF applications

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Abstract

In-situ resources play an important role in future extraterrestrial human activities for facility repair and habitat construction, especially in sustainable space exploration of the Moon and Mars. This paper verifies the feasibility of lunar and Martian regolith simulant for ceramic joining in which SiC ceramic was made into specific size samples and then brazed with the regolith under argon atmosphere, respectively. The microstructure and interfacial phase of the joints were characterized in detail. The results show that it is possible to braze ceramic parts together with extraterrestrial regolith simulant. The average ultimate shear strength of joints of SiC brazed with lunar regolith simulant (LRS-SiC), and SiC brazed with Martian regolith simulant (MRS-SiC) was 21.5 ± 6.5 MPa and 27.3 ± 13.4 MPa, respectively. Finally, the joining mechanism of the bonded joints was proposed. Intergranular diffusion of elements and new phases formed by reactions were identified as the main mechanism of bonding in the interface of SiC/glass. The presented method may provide insight into equipment maintenance associated with repairing and fabricating in subsequent Moon and Mars missions.
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Keywords: ISRU; ISRF; Extraterrestrial regolith; Brazing; SiC ceramic

1. Introduction

Establishing a permanent and self-sufficient safe lunar base on Moon is foreseen to be one of the next steps in human space exploration (Benaroya, 2018). This process would require a diverse range of raw materials for equipment maintenance and infrastructure construction. Launching all resources from Earth is both cost and time-prohibitive using current technology (Costs, 2002; Duke et al., 2003). The *in-situ* resource utilization (ISRU) and *in-situ* fabrication and repair (ISFR) technologies can radically reduce the mass, cost, and risk of given missions, referring to utilizing local resources for the fabrication and repair of architectural structures and equipment components (Bassler et al., 2006; Hammond et al., 2006; Howell et al., 2008; Naser, 2019; Sanders and Larson, 2012).

At present, physical asset fabrication from extraterrestrial regolith covering the surface of the planets is one of the main research objectives. With cold treatment of the regolith, it was fabricated into the geopolymers, polymer composites, and concrete for habitats construction (Altun et al. 2021; Buchner et al., 2018; Cai et al., 2019; Cesaretti et al., 2014; Davis et al., 2017; Wang et al., 2017; Reches, 2019). High-temperature bonding of the regolith was also studied by researchers due to the simple process which only requires heat and the material itself. The regolith undergoes sintering and melting by selective laser sintering (SLS) methods (Fateri & Gebhardt, 2015; Gerdes et al., 2018; Goulas et al., 2019; Xu et al., 2019)

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Table 1
Chemical compositions of HIT-L-1 LRS and HIT-M-1 MRS (%, mass fraction).

	HIT-L-1	HIT-M-1
SiO_2	39.16	43.20
Al_2O_3	13.41	15.29
CaO	17.10	9.87
FeO_t	17.81	18.50
MgO	2.45	2.82
TiO_2	3.25	3.17
Na_2O	1.88	2.49
Cr_2O_3	0.02	0.02
MnO	0.37	0.37
K_2O	2.71	2.52
P_2O_5	0.29	0.55
SO_3	0.02	0.02
Cl	0.04	0.04
LOI	0.72	0.58
SUM	99.21	99.34

and solar energy sintering methods (Fateri et al., 2019a; Imhof et al., 2017; Meurisse et al., 2018; Urbina et al., 2017). Generally, such processes integrate with additive manufacturing technologies to promote the feasibility of implementation in space environments.

Space missions also face *in-situ* maintenance problems. NASA's Curiosity Rover, which has been active on Mars for more than 9 years since its landing in August 2012, suffered from unexpected considerable wheel damage beginning in October 2013 (Toupet et al., 2018). Aiming at the repair demands of metallic materials, we proposed a fusion welding scheme directly by extraterrestrial regolith and solar energy (Zheng and Qiao, 2020). In addition to metallic materials, ceramics are an equally important material for aerospace applications in aircraft, spacecraft, and launch vehicles due to their high-temperature capability, stiffness and strengths, and excellent resistance to oxidation and corrosion. Currently used engineering and structural ceramics in aerospace include ceramic thermal and environmental barrier coatings for protecting hot-section components of aircraft turbine engines from high heat flux in high-temperature combustion environments, rocket exhaust nozzles, and thermal protection systems for space vehicles (Barsoum, 2019; Bhat, 2018). On the extraterrestrial regolith interreacting with ceramics, Dominguez and

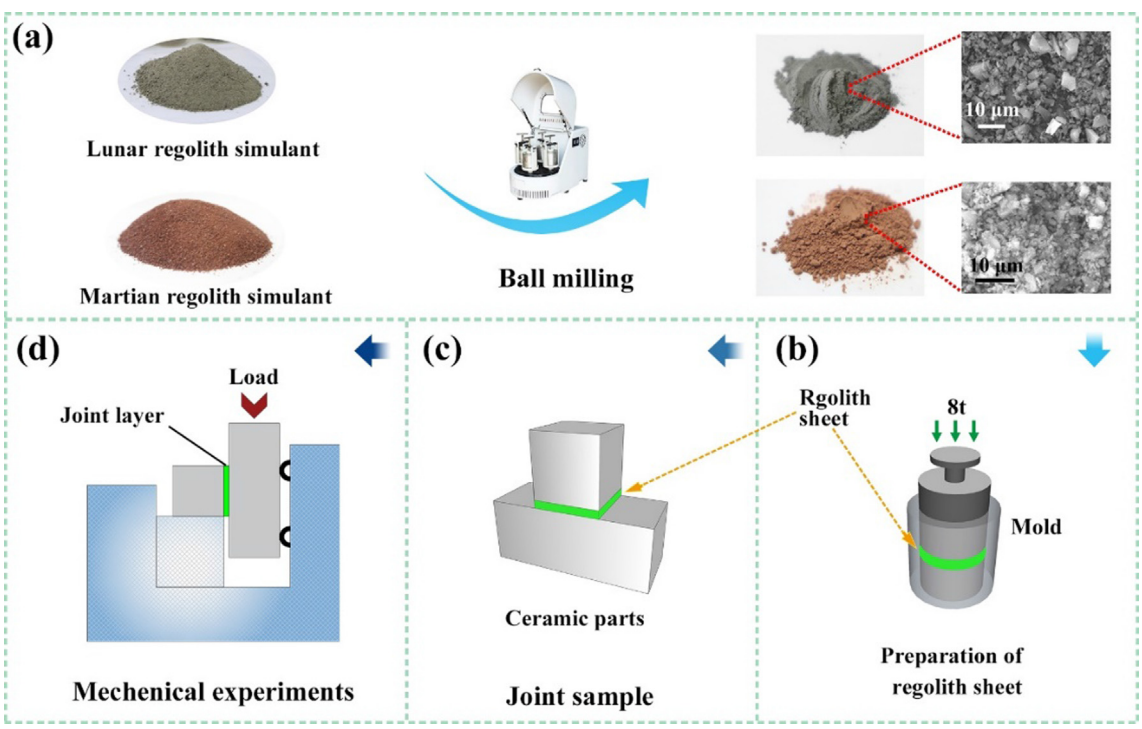

Fig. 1. The schematic description of the jointing procedures: (a) the preparation of HIT-L-1 and HIT-M-1 filler powder; (b) the preparation of regolith sheet; (c) the assembly of joint sample; (d) schematic of rig used for shear testing.

Table 2
Components and mechanical properties of SiC ceramic employed in this work.

Purity (%)	Density (g/cm^3)	Flexural Strength (MPa)	Elastic modulus (GPa)	Thermal conductivity (W/m.k)	Fracture toughness (MPa.m½)
98.5	3.12	400	415	148	4.5

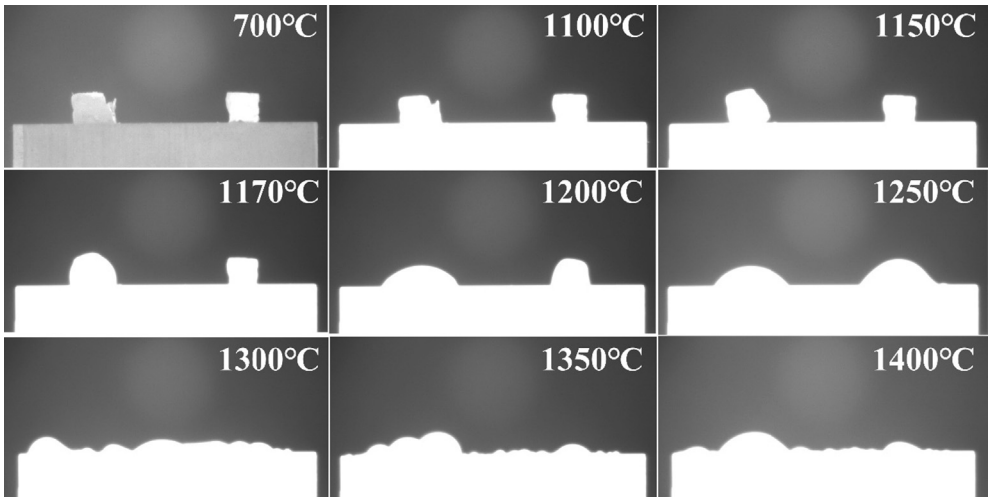

Fig. 2. Wetting properties of HIT-L-1 LRS (left) and HIT-M-1 MRS (right) on SiC substrates at a temperature range of 700 °C–1400 °C.

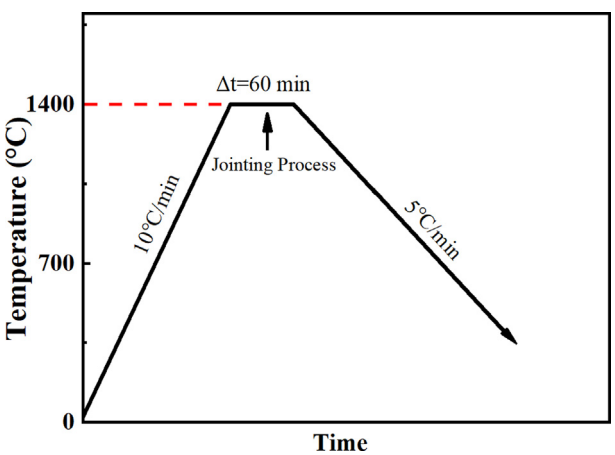

Fig. 3. The temperature control curves of brazing experiments.

Whitlow (2019) observed the upwards migration phenomenon on molten lunar regolith as melting the lunar regolith within alumina crucibles under 10^{-2} torr. Fateri et al., 2019b) investigated the wettability of the melt lunar regolith using pre-sintered regolith platforms at a range of 25–1450 °C. Zocca et al. (2020) investigated the wetting behavior of lunar regolith on the alumina substrate. From the results of the above studies, we can infer that extraterrestrial regolith has a great potential for solving the joining and repair demands of ceramic materials in future aerospace applications.

Silicon carbide (SiC) ceramic and related materials have been widely used in different military and engineering fields, such as aerospace, armor, space mirror, nuclear, chemical industry owing to their unique promising properties of low density, high mechanical strength, good thermal stability, and excellent chemical stability (Chen et al., 2020; Eom et al., 2013; Liu et al., 2020). Moreover, SiC ceramic belongs to the third generation of semiconductor materials, which has a wide bandgap, high breakdown electric field, high thermal conductivity, high electron saturation rate, and higher radiation resistance. Therefore, SiC components joining and repairing would likely become a critical problem for sustainable long-term space equipment maintenance on Moon and Mars in the future. In practical engineering, brazing is the main way of joining ceramics, in which molten filler materials flow into the joint. Herein, we investigate the feasibility of lunar and Martian regolith for jointing SiC ceramic under Argon conditions. The brazing procedures are introduced in detail, and the mechanical properties of the brazing joints were evaluated by a testing instrument. Additionally, the microstructures of the soldering seam of jointed samples are characterized. This research provides a possible method for utilizing in-situ resources on equipment maintenance involving ceramics joining in future space missions.

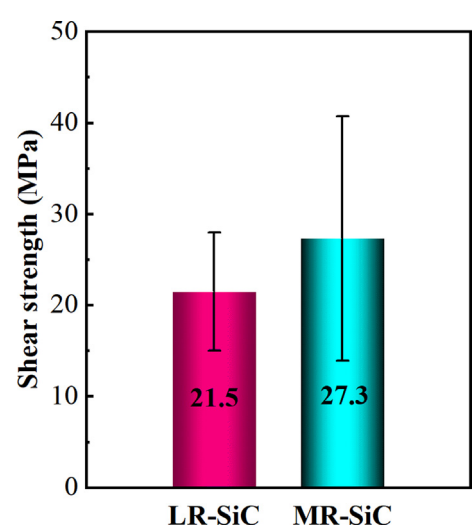

Fig. 4. Shear strength of the two joints at room temperatures.

2. Experimental procedures

2.1. Materials preparation

2.1.1. Extraterrestrial regolith simulant

HIT-L-1 lunar regolith simulant (LRS) and HIT-M-1 Martian regolith simulant (MRS) (Zheng and Qiao, 2020) were employed in this paper, the chemical compositions of which are listed in Table 1. The grinding process is as follows. 100 g regolith simulant (<200 μm) was mixed with 400 g aluminum oxide grinding balls and 40 g alcohol in a 1000 ml corundum bottle. The mixture was ground at the speed of 400 rpm for 2 h in a ball mill machine to obtain the powder with a particle size < 10 μm, as shown in Fig. 1(a). Afterward, the ground simulants were cleaned in an ultrasonic cleaner, and dried at 70 °C for 12 h. 130 mg ball-milled regolith powder was subsequently set into the mold with a chamber size of Φ 13 mm for the starting filler sheet used in the joining process, in which a press machine gives an 8-ton pressure on the mold with 2 min holding time, as shown in Fig. 1(b). The mean thickness of LRS and MRS sheets was 1047.346 ± 18.094 μm and 934.327 ± 7.194 μm, respectively, which was observed by an Opto-digital microscope (DSX500, OLYMPUS Co., JPN).

2.1.2. Ceramics

The pressureless sintering SiC ceramics (Guangzhou Kaihong Electronic Technology Co., Ltd) was employed for the demonstration of the brazing process, whose profiles are listed in Table 2. The ceramic blocks were cut by a diamond-cutting machine into two kinds of dimensions, $4 \times 4 \times 4$ mm^3 and $4 \times 4 \times 10$ mm^3, and the faying faces of the substrates were ground on a 1500-grit diamond disc before the joining experiments.

2.2. Determination of joining temperature and holding times

The wetting behavior of HIT-L-1 LRS and HIT-M-1 MRS on SiC substrates was investigated to determine an appropriate temperature for the brazing process. Herein,

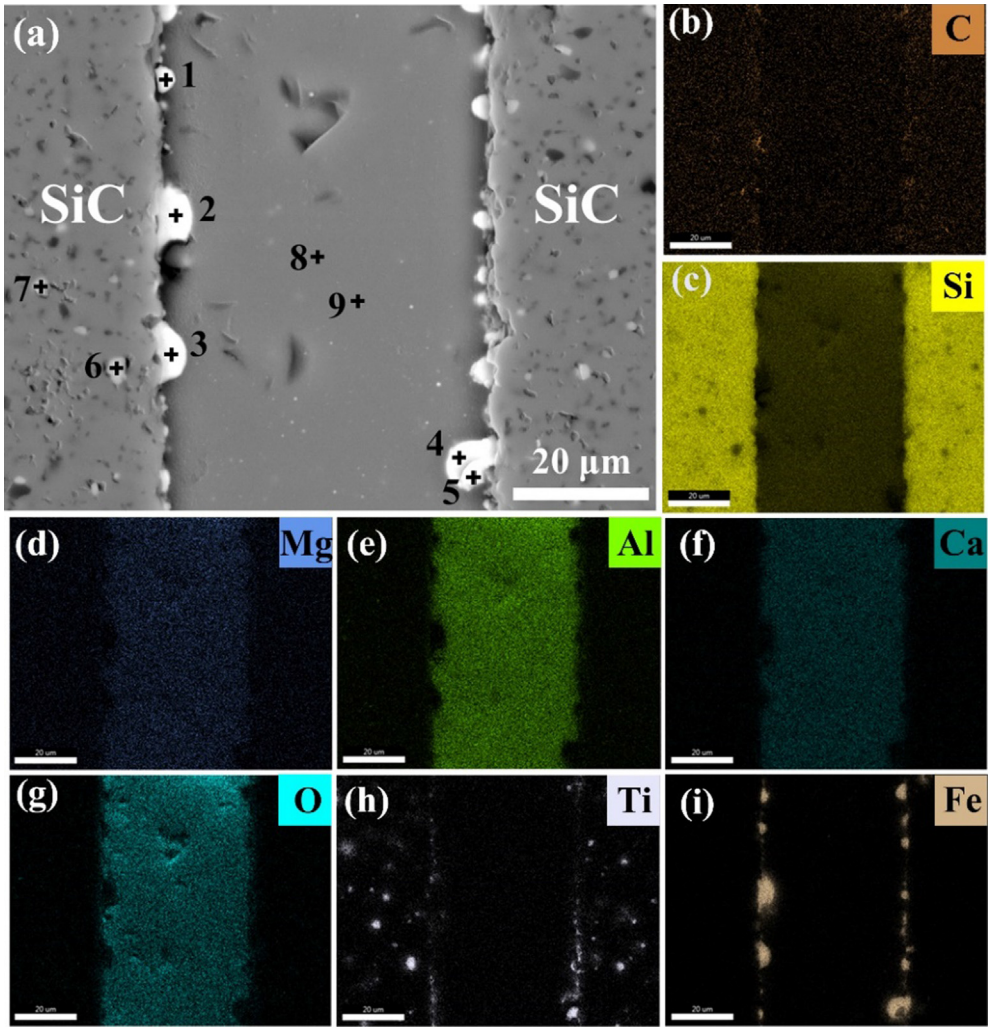

Fig. 5. Backscattered electron images of the SiC joint brazed by HIT-L-1 filler at 1400 °C for 60 min: (a) micrograph of the joint; (b-i) mapping results of elements distribution of the joint.

Table 3
EDS results of atomic composition in points 1–9 marked in Fig. 5(a).

Points	Elements (at.%)									
	C	O	Na	Mg	Al	Si	K	Ca	Ti	Fe
1	26.35	16.9	0.24	0.93	3.40	26.06	0.12	0.48	1.01	24.50
2	27.59	8.14	0.17	0.38	1.05	22.09	0.12	0.18	0.30	39.97
3	28.12	6.93	0.26	0.71	2.42	23.06	0.12	0.36	0.17	37.86
4	44.29	8.27	0.19	0.47	1.29	11.75	0.08	0.15	17.82	15.69
5	30.34	14.33	0.29	1.04	3.14	21.95	0.12	0.32	1.00	27.47
6	43.43	2.81	0.66	0.66	0.96	33.21	0.03	0.09	17.97	0.18
7	34.61	2.61	0.71	0.81	1.12	46.08	0.05	0.11	13.73	0.17
8	7.97	54.07	0.31	2.51	9.89	22.32	0.32	2.40	0.08	0.14
9	9.19	53.17	0.17	2.47	9.80	22.31	0.33	2.37	0.08	0.11

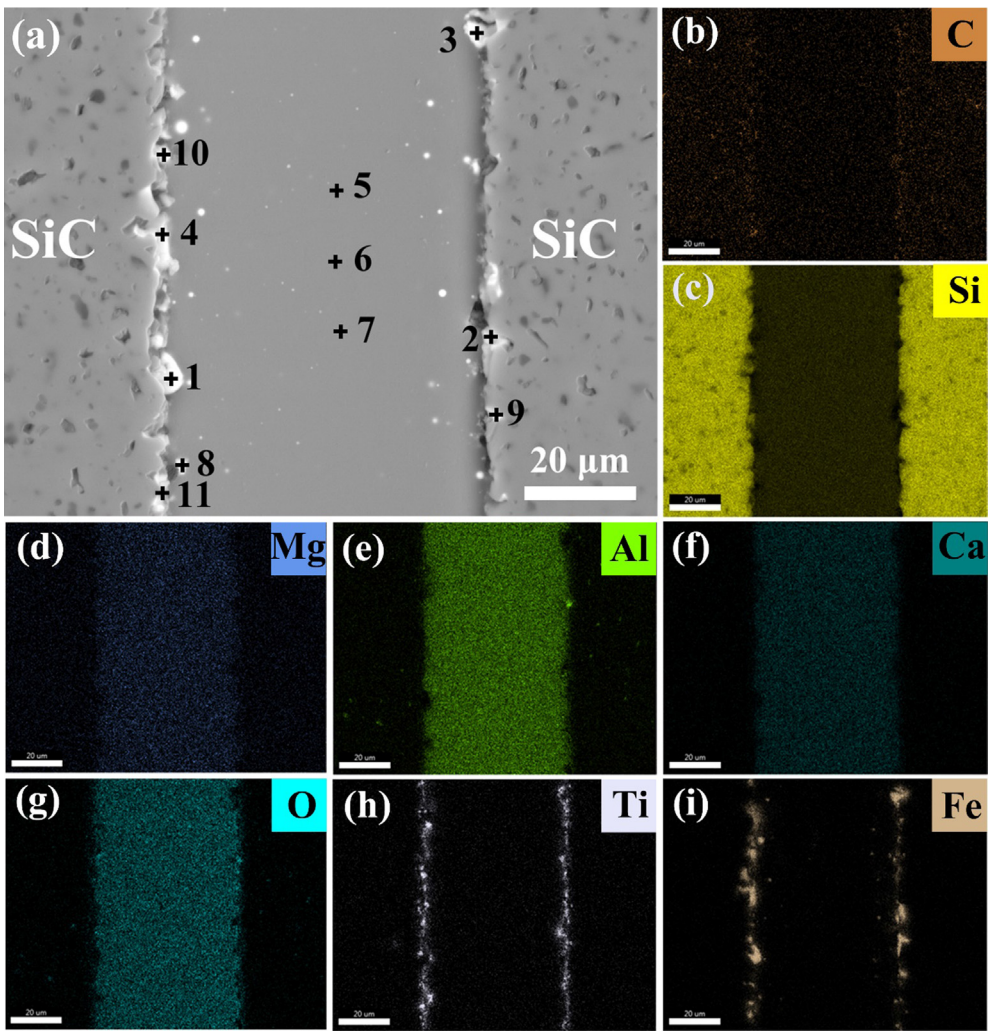

Fig. 6. Backscattered electron images of the SiC joint brazed by HIT-M-1 filler at 1400 °C for 60 min: (a) micrograph of the joint; (b-i) mapping results of elements distribution of the joint.

the sessile drop method was carried out at a heating rate of 5 °C/min in a tube furnace equipped with a camera (DV-HV 1303UM, CHN) from room temperature (RT) to 1400 °C under an argon atmosphere. The sample's geometry during the heating process was recorded by a computerized image analysis system. The results were as shown in Fig. 2, and the fully melting temperature of the two simulants happened before 1400 °C which was taken as the joining temperature for the brazing process. Additionally, the furnace temperature was held after it reached 1400 °C, during which the sample's geometry changes via time were recorded (please see supplementary movie). It can

Table 4
EDS results of atomic composition in points 1–11 marked in Fig. 6(a).

Points	Elements (at.%)									
	C	O	Na	Mg	Al	Si	K	Ca	Ti	Fe
1	15.57	9.06	0.48	1.06	3.58	21.16	0.3	0.47	1.98	46.34
2	10.98	10.00	0.71	1.62	4.60	24.84	0.24	0.52	2.32	44.19
3	22.22	5.56	0.17	0.81	2.47	27.60	0.20	0.41	1.85	38.7
4	17.98	18.27	0.47	1.72	6.08	22.4	0.25	0.59	1.40	30.85
5	0.05	51.70	0.45	3.46	13.45	26.19	0.84	3.59	0.12	0.16
6	2.80	50.67	0.43	3.44	13.10	25.03	0.78	3.44	0.14	0.17
7	3.74	49.97	0.47	3.35	12.92	25.01	0.87	3.42	0.11	0.14
8	17.41	39.48	0.63	2.40	8.73	17.96	0.48	1.65	9.48	1.77
9	28.08	7.87	0.28	0.97	2.45	38.15	0.12	0.25	0.31	21.51
10	22.75	12.45	0.68	1.23	4.12	23.57	0.23	0.46	1.09	33.41
11	27.08	24.79	0.29	1.12	3.61	23.97	0.2	0.50	12.43	6.01

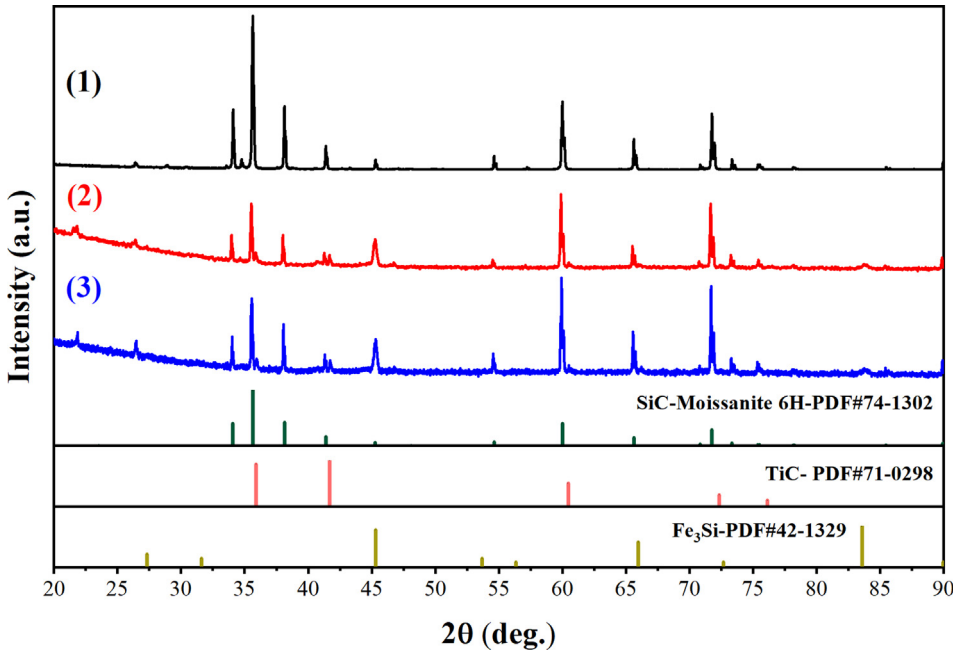

Fig. 7. XRD and corresponding PDF standard patterns for (a) SiC ceramic; (b) joints of LRS-SiC sample; (c) joints of MRS-SiC sample.

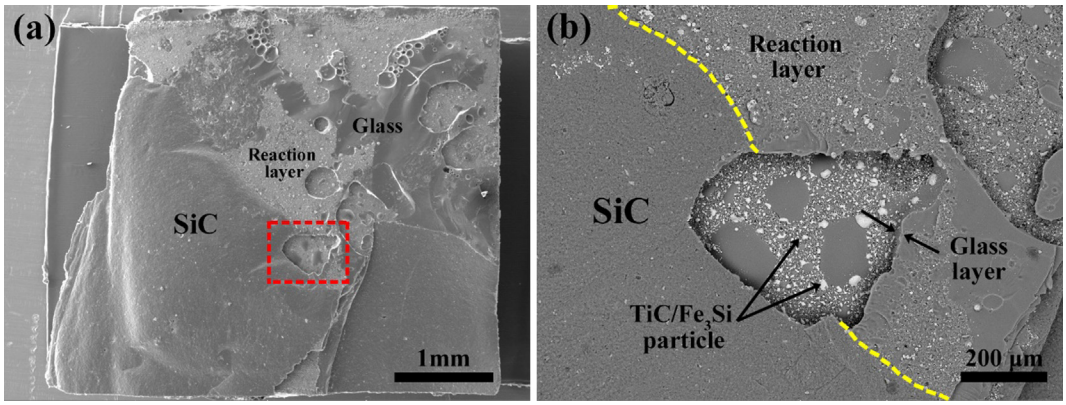

Fig. 8. (a) SEM image of fracture micrographs of the joints after mechanical tests. (b) BSE image of magnification of area marked by the red line of dashes in Fig. 8(a).

be seen that the geometry was approximately the same after 45 min. Herein, we choose 1 h as the holding time for a successful brazing process.

2.3. Brazing experiments

The regolith sheets were evenly infiltrated using ethyl cyanoacrylate adhesive and made into starting fillers. Then the starting filler was put between the parent materials to form a sandwiched brazing couple, as shown in Fig. 1(c). The joining process was executed in a muffle furnace under Argon atmosphere. In the beginning, the samples were heated to 1400 °C with a 10 °C/min heating rate, followed by a holding stage that lasted for 60 min. Finally, the specimens were cooled to room temperature at a low rate of 5 °C/min. In practice, the cyanoacrylate adhesive with a boiling point of 200 °C fully volatilizes before 400 °C at the furnace, ensuring it has little influence on the brazing process. The temperature control curves of brazing experiments were as shown in Fig. 3.

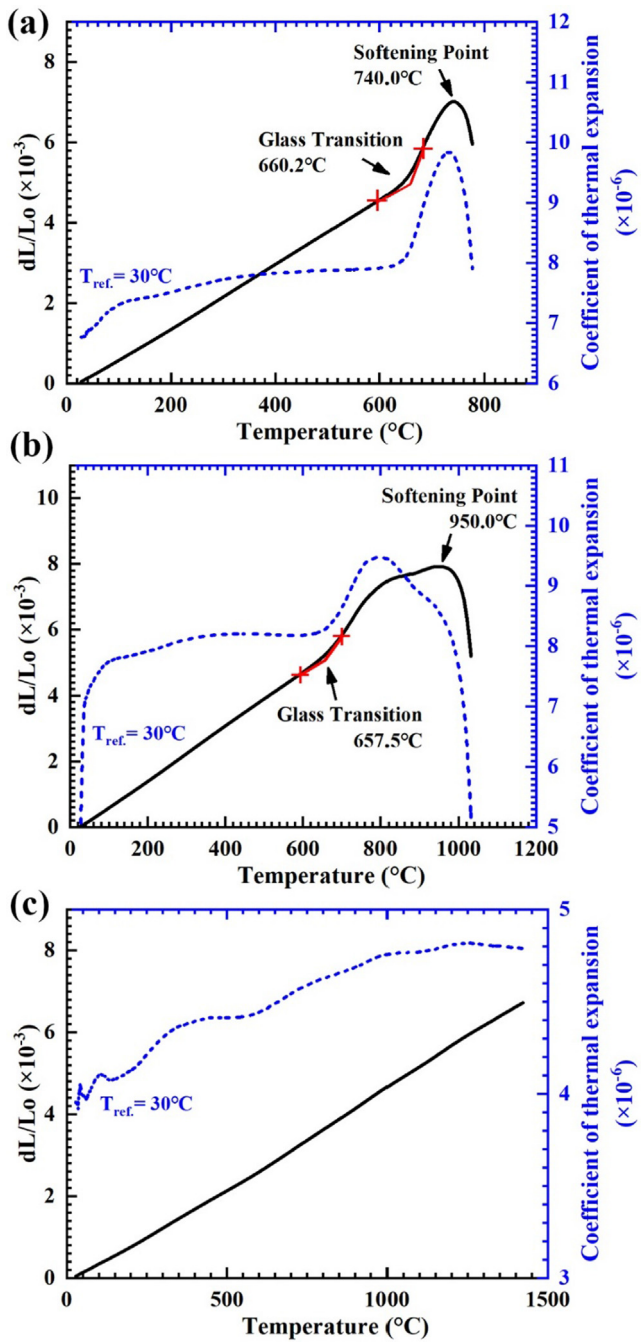

Fig. 9. The CTE curves of the materials used in the work: (a) LRS glass; (b) MRS glass; (c) SiC ceramic.

2.4. Thermal expansion experiments

Thermal compatibility between SiC ceramic and the filler layer plays an important role in joint strength during the brazing process. We measured the mean linear coefficient of thermal expansion (CTE) of SiC ceramic and the filler layer materials using a dilatometer (DIL 402C, Netzsch, DEU). In correspondence with the same conditions as the brazing process, the filler materials of LRS and MRS were fabricated using the ingot casting method (Zheng and Qiao, 2022) and treated with the same temperature control curves in Fig. 3. The obtained glass of LRS and MRS along with the SiC ceramic was then cut into the size of 4 × 4 × 25 mm^3 by a diamond-cutting machine for the DIL tests.

2.5. Characterization methods

The shear tests of joints were evaluated at room temperature using a test machine (Instron 5569, USA) at a loading speed of 0.2 mm/min. The schematic of the shear strength test of the joints is shown in Fig. 1(d). The microstructure of the joints was investigated by scanning electron microscopy (SEM, Vega3, Czech) which is equipped with an energy dispersive spectrometer (EDS, Element, USA). In addition, crystalline phases in the joints were further identified by X-ray diffraction (XRD, Panalytical, NLD).

3. Results and discussion

3.1. Shear strength of brazing samples

In brazing experiments, two brazing forms including SiC ceramics joined with HIT-L-1 lunar regolith simulant (LRS-SiC) and SiC ceramics joined with HIT-M-1 Mars regolith simulant (MRS-SiC) were deployed. Seven parallels were taken for each form, making 14 joined specimens. The average ultimate shear strength of the two joints at room temperatures is illustrated in Fig. 4, which is 21.5 ± 6.5 MPa for SiC-LRS and 27.3 ± 13.4 MPa for SiC-MRS. Such strength was lower than that of the SiC_f /SiC joint brazed by Y-Al-Si-O glass (38.85 MPa) (Fan et al., 2018), which is a common filler system for joining of SiC (Herrmann et al., 2014). We believe these shear strength numbers could address emergency circumstances of joining and repair of space facilities, such as space mirrors and

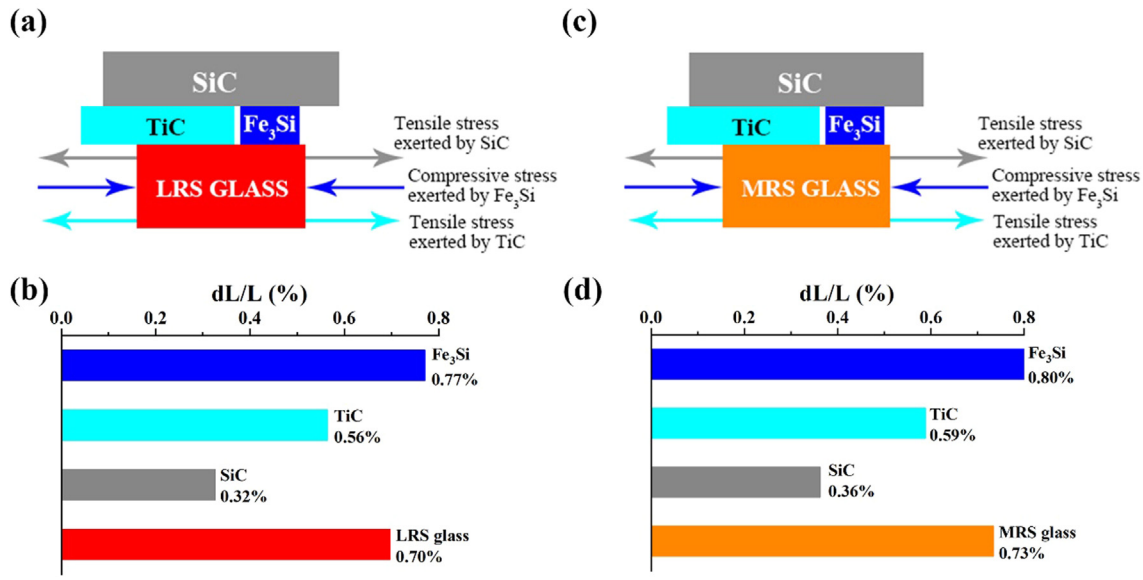

Fig. 10. (a) The stress distributions of LRS glass (730 °C–30 °C). (b) The expansion rate of materials in LRS-SiC (730 °C–30 °C). (c) The stress distributions of MRS glass (800 °C–30 °C). (d) The expansion rate of materials in MRS-SiC (800 °C–30 °C).

radar shields under extreme conditions on the Moon and Mars. Among seven parallels at each brazing form, one sample was selected for discussing the microstructures of the joint.

3.2. Microstructures of the brazing joints

Fig. 5a shows the backscattered electronic (BSE) images of the SiC joints brazed using HIT-L-1 filler (LR-SiC). It can be seen that the SiC substrate has random voids, whereas the filler layer is relatively dense. The thickness of the brazing seam was 46.750 μm. From the elements mapping results [Fig. 5(b-i)], HIT-L-1 filler infiltrated into pores of SiC substrate, which indirectly shows the good wettability between the two materials. Moreover, the macroscopic interlayer between them is the reaction layer consisting of two mixed phases containing Ti and Fe elements according to elements mapping results in Fig. 5(h) and (i). Spectrum investigation was carried out on the jointed layers, and the EDS results of the marked points in Fig. 5(a) are listed in Table 3, indicating that possible phases of the reaction layer of the SiC-LR sample are Ti-C compound, Ti-Si compound, Fe-Si compound, and Fe-C compound.

Concerning the SiC joints brazed using HIT-M-1 filler (MR-SiC), Fig. 6(a) shows the BSE images of the joints of the MR-SiC sample, and Fig. 6(b-i) are elements mapping images of the brazing seam. The thickness of the brazing seam was 55.676 μm. It can be seen that Ti and Fe elements concentrated around the interface between SiC and glass filler, meanwhile, all elements infiltrated into the pores of SiC as the LR-SiC case. EDS results of the marked points in Fig. 6(a) are listed in Table 4, showing the possible phases of the reaction layer are the same as that of the LR-SiC sample.

According to the XRD results (shown in Fig. 7), it can be determined that the interlayer consisted of TiC and Fe_3Si, no raw HIT-L-1 or HIT-M-1 phases were detected, which means that all the raw powders took part in the reactions to form glass and the above phases during the bonding process.

3.3. The chemical reactions during the brazing process

From microstructure identification, it can be seen that TiC and Fe_3Si are transition phases between the SiC substrate and glass bonding layer in the brazing seam. The reaction equations are as follows:

$$\mathrm{Ti} + \mathrm{C} = \mathrm{Ti}, \Delta G = -180403.5\mathrm{J},\ 1400\,\mathrm{C} \quad (1)$$

$$3\mathrm{Fe} + \mathrm{Si} = \mathrm{Fe_3Si},\ \Delta G = -74912.8\mathrm{J},\ 1400\,\mathrm{C} \quad (2)$$

The Gibbs free energy change of the reactions was from *Factsage 8.1* database. The morphologies of fractures [Fig. 8(a)] suggested that the bonding strength derived from the interaction of molten glass and the SiC phase could be higher than the shearing strength of SiC base ceramic. Fig. 8(b) is a magnification BSE image of the area marked by the red line of dashes in Fig. 8(a), showing that TiC/Fe_3Si particles are distributed on the glass layer.

3.4. Thermal expansion of SiC ceramic and the filler layer glass

Fig. 9 shows the mean linear CTE curves of the LRS glass, the MRS glass, and SiC employed in this work, and the reference temperature ($T_{ref.}$) is 30 °C. Typical thermal expansion curves of glass were observed in LRS and MRS samples, which exhibit the glass transition ($T_{g.}$) is 660.2 °C and 657.5 °C, respectively; the softening point is

740.0 °C and 950.0 °C, respectively. The glass transition is the temperature at which the amorphous materials transform between the glassy state and the high-elastic state. The maximum CTE of LRS glass, MRS glass is both locates in their high-elastic intervals, which is 9.84×10^{-6} 1/°C (30 ~ 730 °C) and 9.47×10^{-6} 1/°C (30 ~ 800 °C), respectively. At such temperature ranges, the CTE of SiC is 4.57×10^{-6} 1/°C (30 ~ 730 °C) and 4.63×10^{-6} 1/°C (30 ~ 800 °C). Additionally, the CTE of TiC and Fe_3Si previously reported is 8.05×10^{-6} 1/°C (20 ~ 1300 °C) and 11.0×10^{-6} 1/℃ (20 ~ 1070 °C), respectively (Maex, 1995; Song et al., 2002). Based on the above data, we calculate the expansion rate (dL/L_0) and illustrate the stress distributions of LRS glass and MRS glass during the cooling stage of the brazing processes, as shown in Fig. 10. The big difference in CTE between the base ceramic and the filler materials did not lead to the failure brazing, and we believe it is due to a moderate effect attributed by the reaction layers consisting of Fe_3Si with relatively high CTE. Therefore, the key lies in the thermal compatibility by reducing the CTE of filler layer materials for joint strength improvement, which could be realized by crystallization control of the glass layer.

4. Conclusions

This study contributes to verifying the feasibility of brazing SiC ceramic with extraterrestrial regolith simulant, harnessing its potential for joining ceramic in ISRU and ISRF fields on Moon and Mars. The brazing craft is detailed, and the joints of specimens are characterized in terms of mechanical properties and microstructure. The results show that it is possible to braze ceramic parts together with extraterrestrial regolith simulant. The joining mechanism is molten glass derived from the regolith infiltrates into pores of SiC ceramic, during which the new reaction layer consisting of TiC and Fe_3Si by the direct atomic bonding formed by the interaction of molten glass and intergranular phase of SiC. The average ultimate shear strength of SiC brazed with lunar regolith simulant (LRS-SiC), and SiC brazed with Martian regolith simulant (MRS-SiC) was 21.5 ± 6.5 MPa and 27.3 ± 13.4 MPa, respectively. This in-situ joining method could be used for equipment repair associated with ceramic materials for future Moon and Mars missions.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.asr.2022.05.015.

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